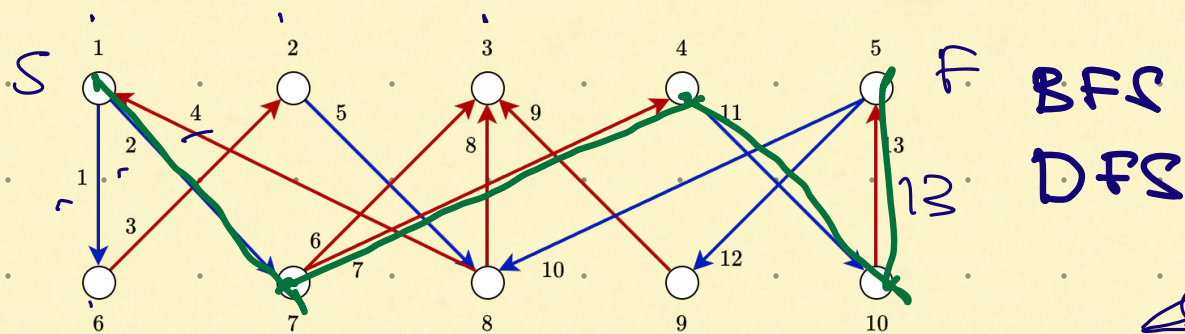




$$Q(r; \underline{\sigma}) = \sum_{w \in \text{Inc}(r)} \sigma_w + q_0(r) \sigma_0$$

BFS
DFS

$$\underline{\sigma} \rightarrow \underline{\sigma}' : Q(r; \underline{\sigma}') = 0 \text{ for all } r$$

$$\text{minimum } \sum_r Q^2(r; \sigma') = 0.$$

Problem " $Q(\underline{\sigma}) = 0$,

what does this mean?

(a) Traditional Relegation $\rightarrow Q(\underline{\sigma}) = \infty$.

$\hookrightarrow$ solving $Q(\underline{\sigma}) = \infty$,

may be approached as a path problem

$\hookrightarrow$ BFS, DFS, ...

(b) Unconventional

Suppose we have direct means
for $Q(\sigma) = 0$.

Model of computation

Problem $\rightarrow$ path $Q(t) = 0 \rightarrow$ Backend,
 $\xleftarrow{\text{unconventional computing.}}$

(c) What kind of problems?

Path, Fair distribution, Discrete tomography,
Quantum encoding, ...

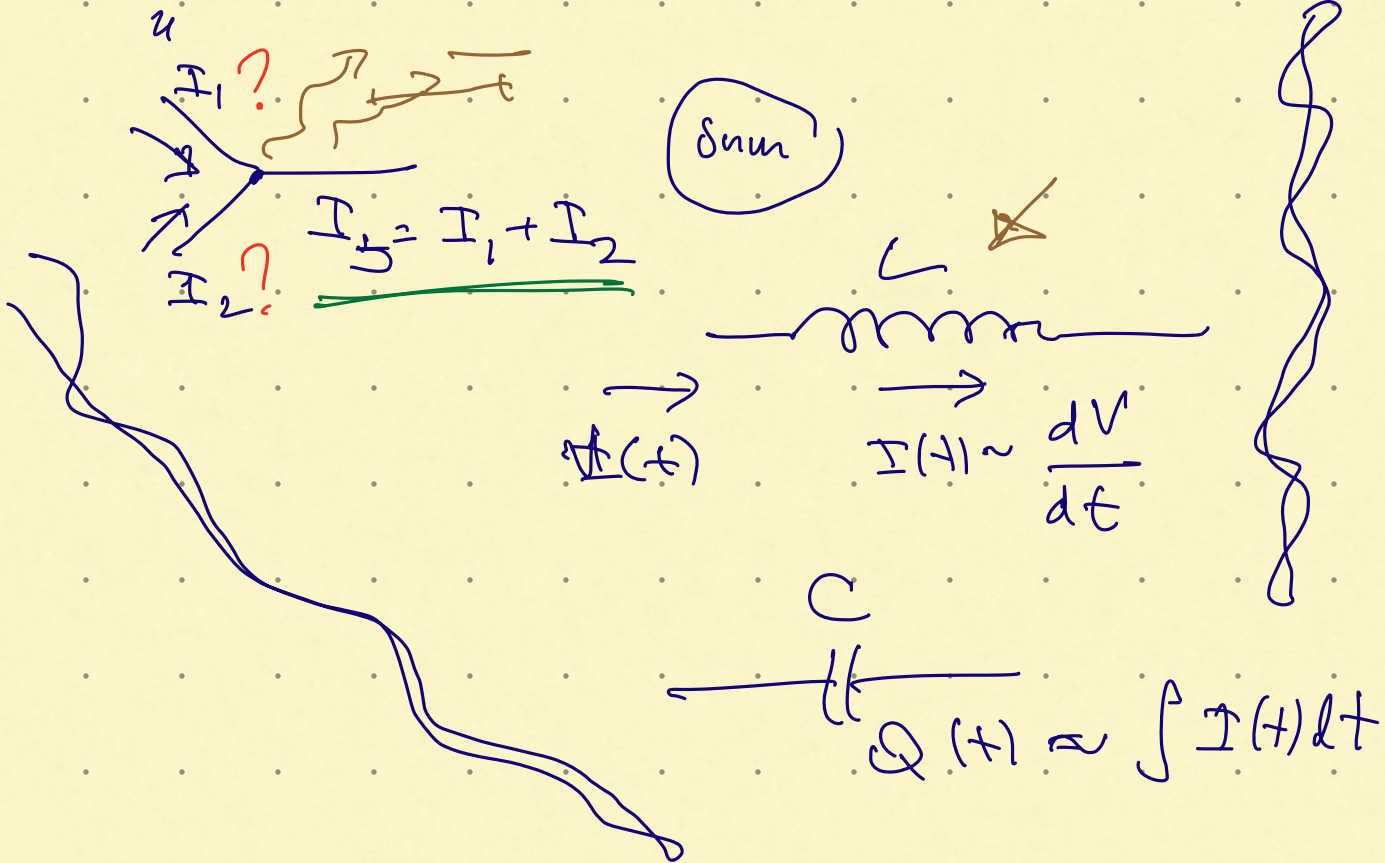
Formal: (any)

Analog computing.

Shannon UAC.

AC were better than DC.

"Transmission system in modern cars"

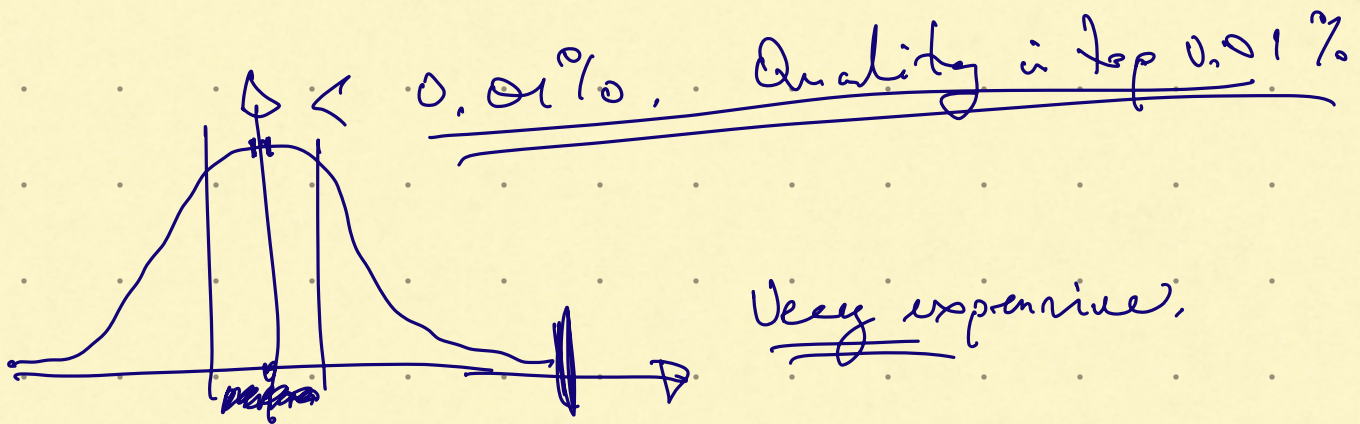


PVT

Errors in devices

→ Errors in computation

Err. Com $< 0.1\%$ $\Rightarrow$ Quality of the device
unit be in
top 0.1% .



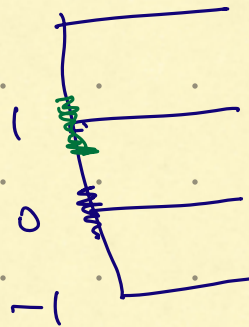
Precision = bits.

Float 32
64
128



Straightforward AC is not scalable

$Q(\underline{x}) = 0$



- ? may scale
logarithmically.

Abandon direct correspondence
for complex problems.

$Q(\sigma) = 0$ can kind of be solved.

V_2 model.

$$\sigma^* = \arg\min_{\underline{\sigma}} Q^2(\sigma)$$

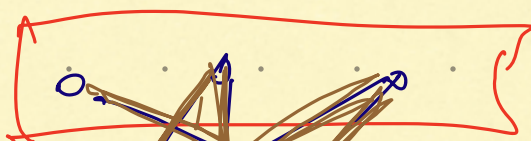
Ising Hamiltonian

$$\sigma^* = \arg\min_{\underline{\sigma}} H(\sigma)$$

ground state.

$$\sigma^* = \arg\max_{\underline{\sigma}} Q(\sigma)$$

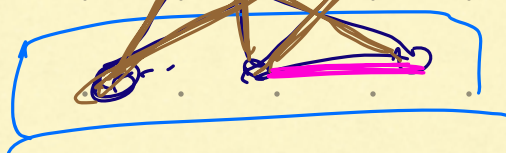
I



$$u \in V$$

$$e \in E$$

II



$$e = (u, v)$$

$$A_{u,v} = \text{weight}(u, v)$$

Cut weight = Total weight of cut

$$\sigma_u = \begin{cases} 1, & \text{if } u \in \underline{I} \\ -1, & \text{if } u \in \underline{II} \end{cases}$$

$$\frac{1}{2}(1 - \sigma_u \sigma_v) = \begin{cases} 0, & \text{if } (u, v) \text{ is not cut} \\ 1, & \text{if } (u, v) \text{ is cut.} \end{cases}$$

$$C(\underline{\sigma}) = \frac{1}{2} \sum_{m,n} A_{m,n} \frac{1}{2} (1 - \sigma_m \sigma_n)$$

$$= \frac{1}{2} \sum_{m,n} A_{m,n} - \frac{1}{4} \sum_{m,n} A_{m,n} \sigma_m \sigma_n$$

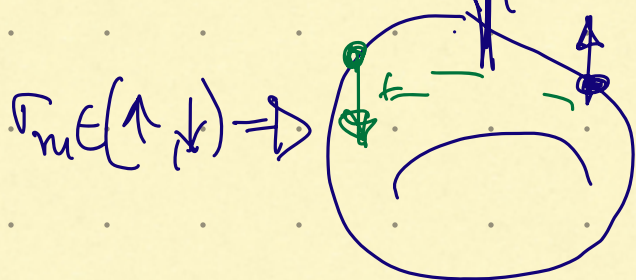
$$C(\underline{\sigma}) = \frac{1}{2} \sum_{m,n} A_{m,n} - \frac{1}{2} H(\underline{\sigma})$$

$$H(\underline{\sigma}) = \frac{1}{2} \sum_{m,n} A_{m,n} \sigma_m \sigma_n$$

Relaxation-Based dyn. Ising Machines

100 spins, $2^{100} = 2^{30} \cdot 10^{10}$ y.s.
 10^{20} words

100 edges, $10 \times 2 = 20$



$$\psi = (\sigma, x)$$

$$\sigma \in \{-1, 1\}$$

$$x \in [-1, 1]$$

+ transition

Definition:

$$V_2: \quad \underline{C(\sigma, x)} = \underline{C(\sigma)} + \Delta C_{V_2}(\sigma, x)$$

$$\Delta C_{V_2}(\sigma, x) = \frac{1}{4} \sum_{m,n} A_{m,n} \sigma_m \sigma_n \underline{|x_m - x_n|}$$

$$x_m = 2 \frac{\partial C(\sigma, x)}{\partial x_m}$$

$$= \sigma_m \sum_n \underline{A_{m,n}} \sigma_n \text{sign}(x_m - x_n)$$

$$\psi(\sigma) = \text{RND}(\sigma) \xrightarrow{T} \underline{Q(\sigma) = 0},$$

RND(x) "

NP-hard problems $\Rightarrow P \sim e^{-N}$

non hard

$$\Rightarrow P \sim \frac{1}{\sqrt{N}}, \left(\frac{1}{\sqrt{N}} \right)$$

Ring oscillators $\rightarrow$ R-DLM